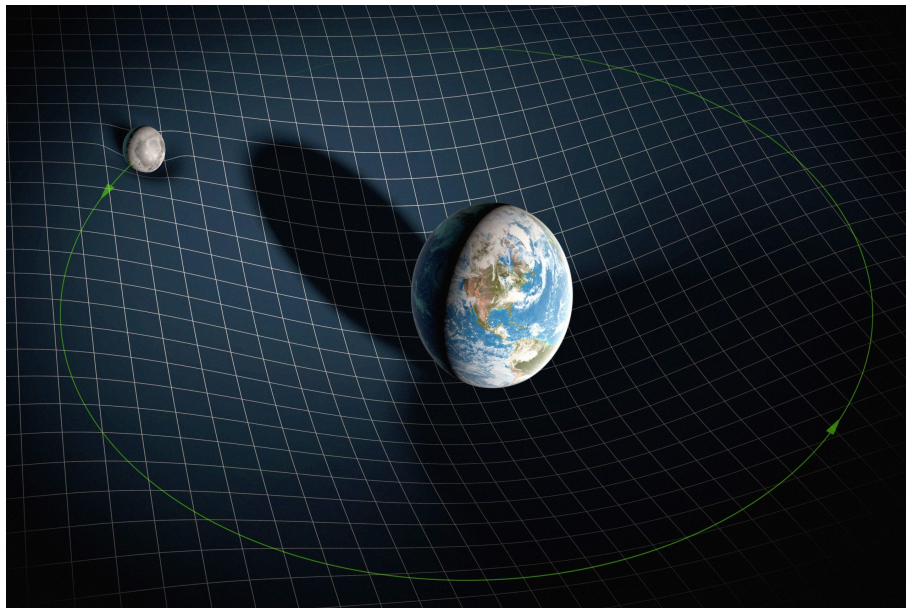




Faculty Of Sciences

Dynamical analysis of a dark-matter halo and galaxies in cosmological, hydrodynamical simulations.



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Abstract

Keywords:

Chapter 1

Introduction

1.1 Cosmology

1.1.1 Cosmological parameters

$$1 + z = \frac{1}{a} \quad (1.1)$$

$$\Omega_m(z) = \frac{\Omega_{m,0}(1+z)^3}{\Omega_{m,0}(1+z)^3 + \Omega_{\Lambda,0}} \quad (1.2)$$

1.1.2 Unit system

$$kpc, M_{\odot}, Gyr \quad (1.3)$$

A general function can be expanded in spherical harmonics as

$$f(\mathbf{r}) = f(r, \theta, \phi) = \sum_{l=0}^{\infty} \sum_{m=-l}^l f_{lm}(r) Y_{lm}(\theta, \phi) \quad (1.4)$$

The coefficients are given by

$$f_{lm}(r) = \int Y_{lm}^*(\theta, \phi) f(r, \theta, \phi) d\Omega \quad (1.5)$$

where $d\Omega = \sin\theta d\theta d\phi$ is the solid angle element following the orthonormality relation of the spherical basis functions. For a fully spherically symmetric system, only the monopole term in the expansion survives, i.e. $l = 0, m = 0$.

For a discrete sampling of the underlying function, the above expression amounts to a summation over all particles in a shell of radius r_j .

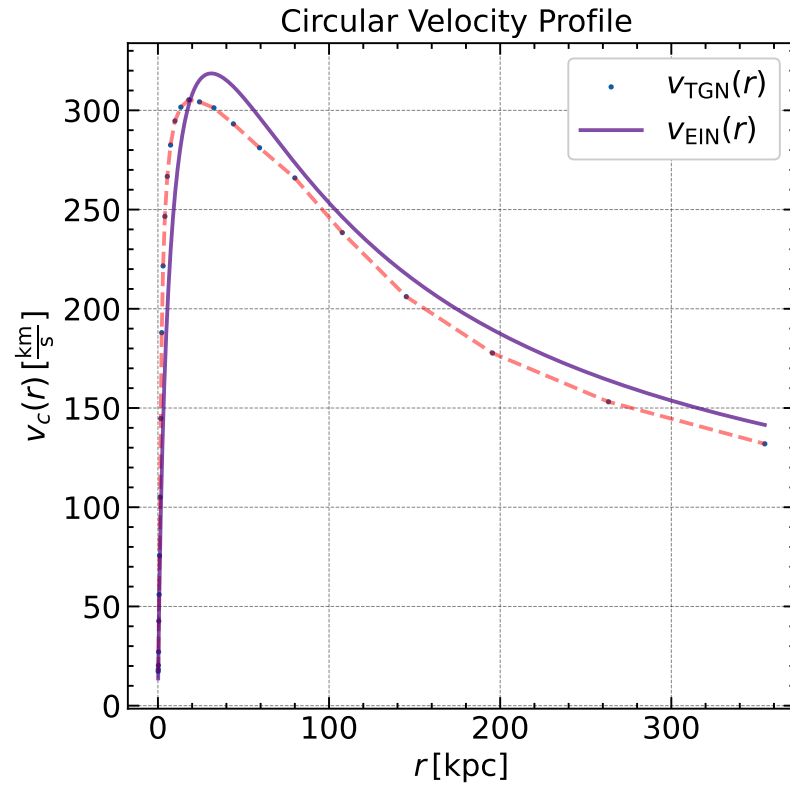
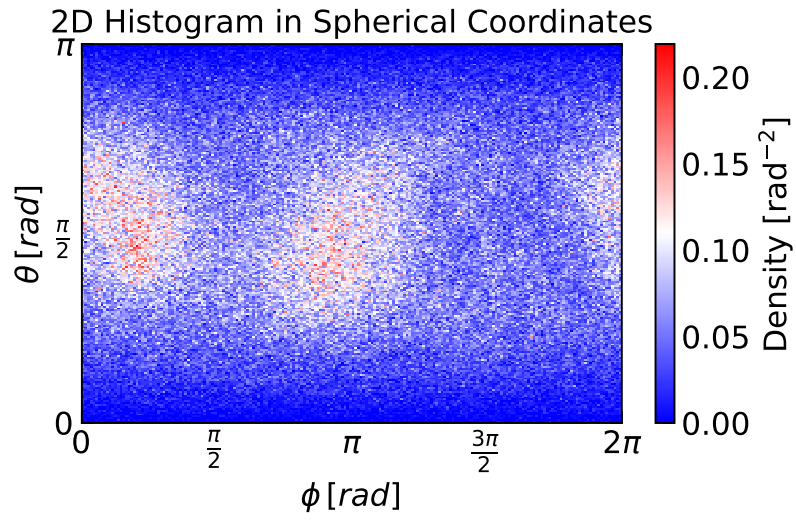
$$f_{lm}(r_j) = \sum_i Y_{lm}^*(\theta_i, \phi_i) f(r_i, \theta_i, \phi_i) \quad (1.6)$$

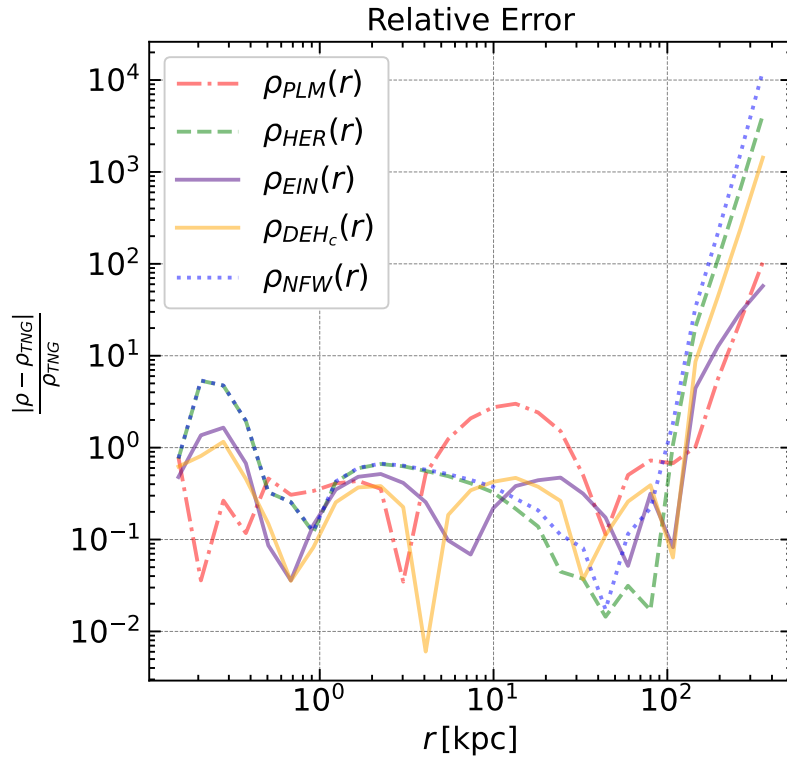
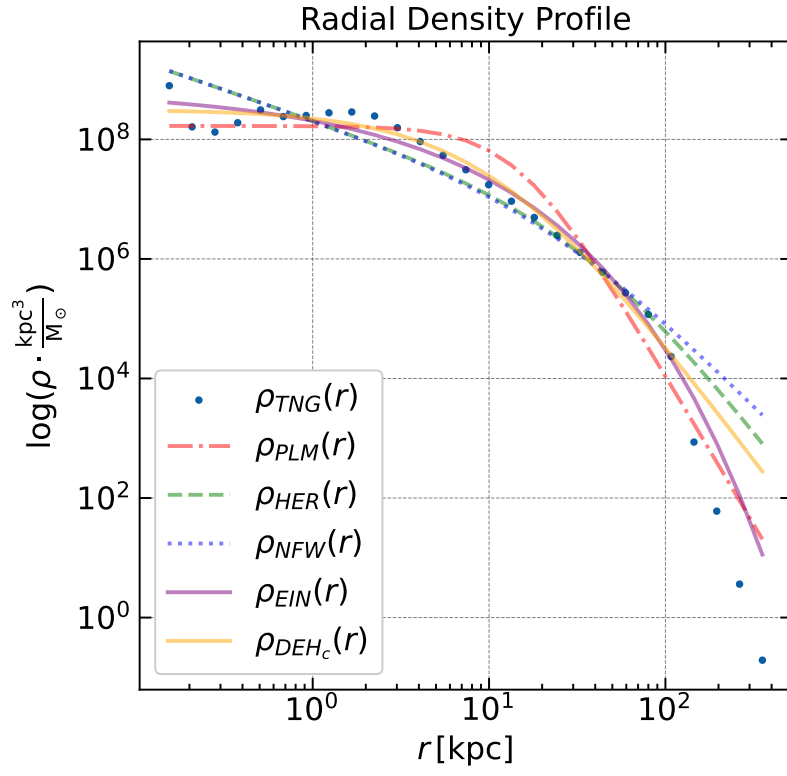
For a discrete distribution of particles, with $\rho(\mathbf{r}_j) = \sum_i \delta(\mathbf{r}_j - \mathbf{r}_i)$, the coefficients can be expressed as

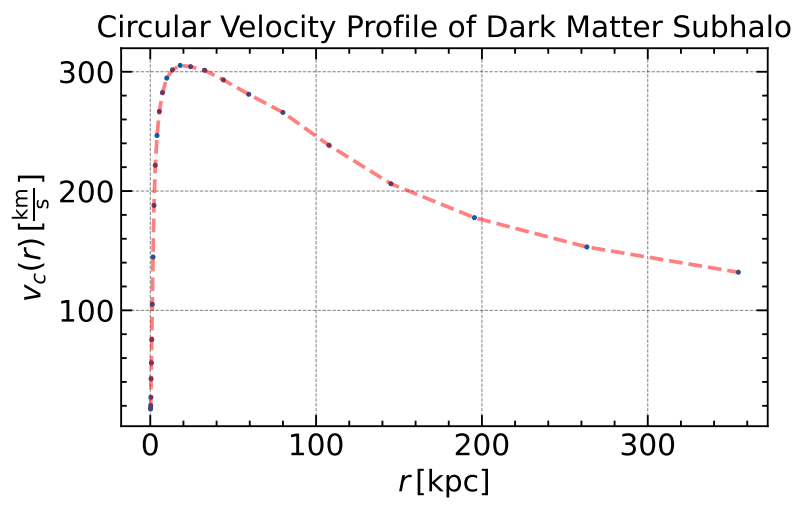
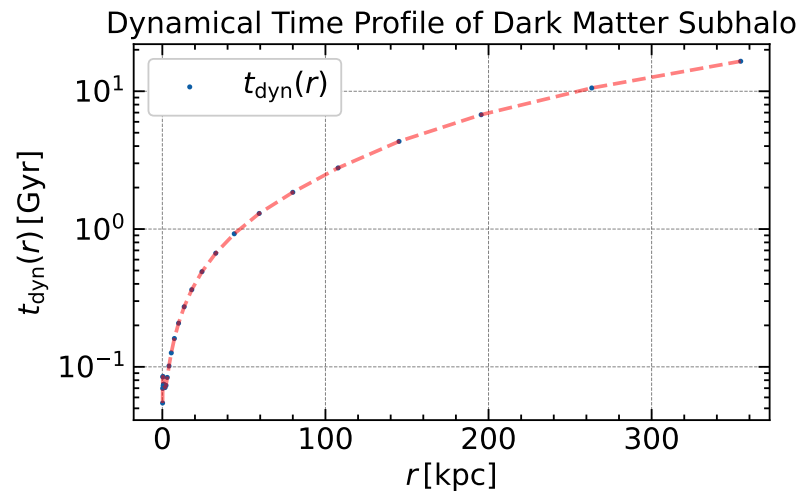
$$\rho_{lm}(r_j) = \int Y_{lm}^*(\theta, \phi) \rho(r_j, \theta, \phi) d\Omega \quad (1.7)$$

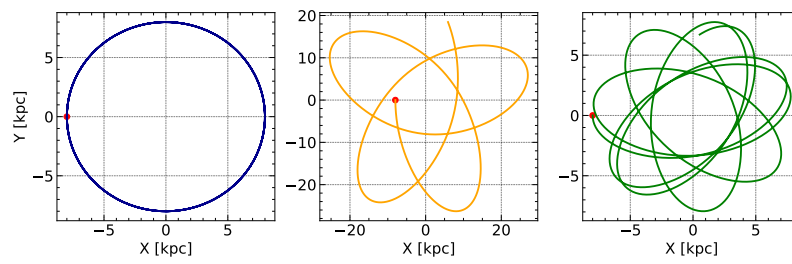
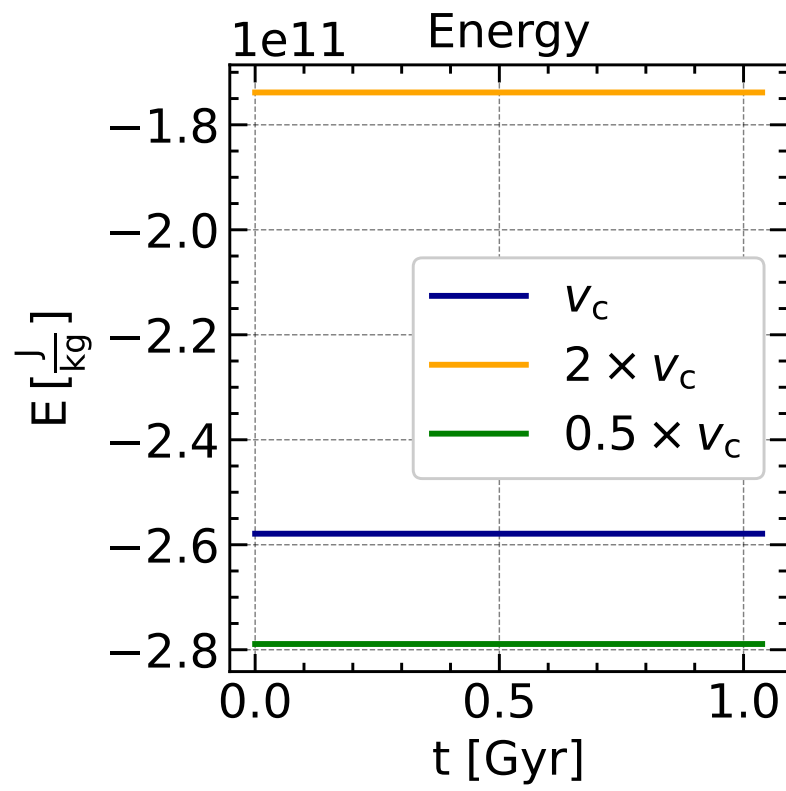
Chapter 2

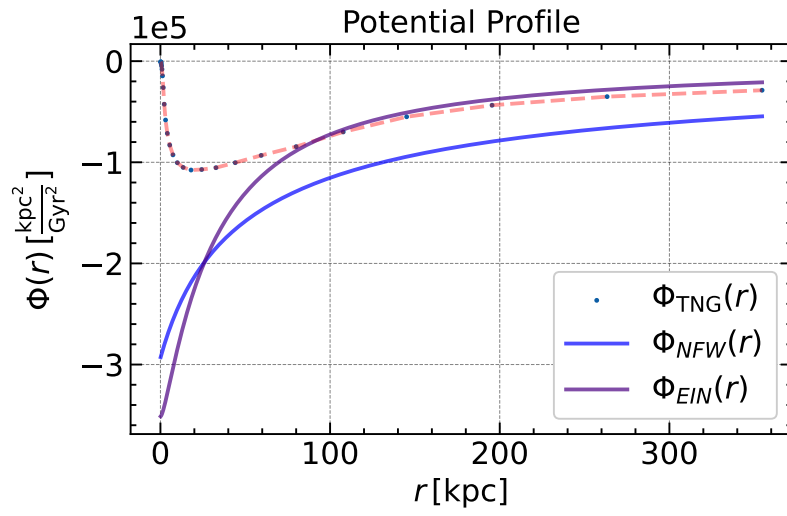
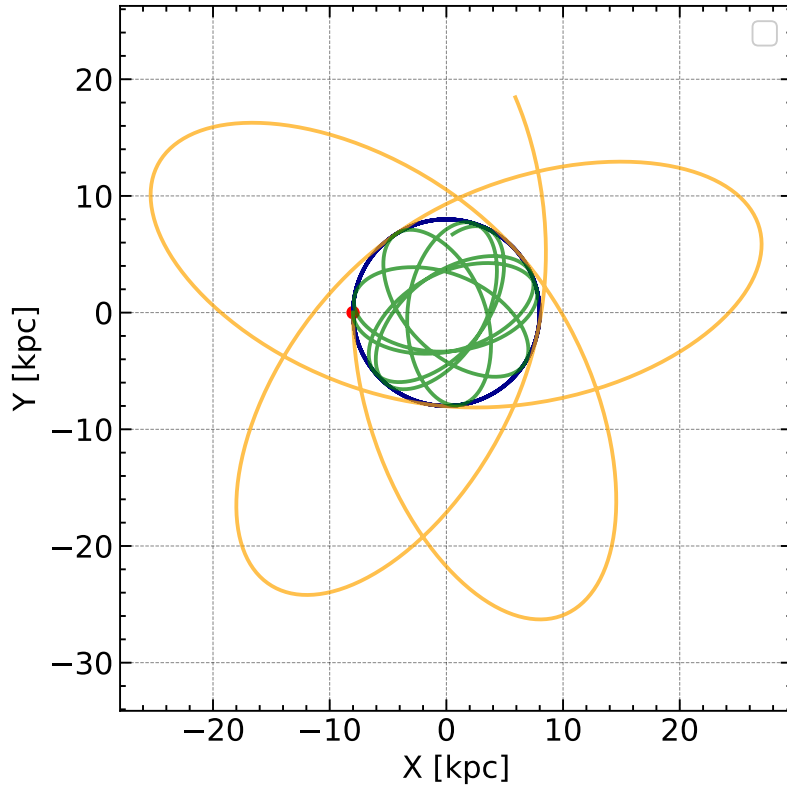
Methodology

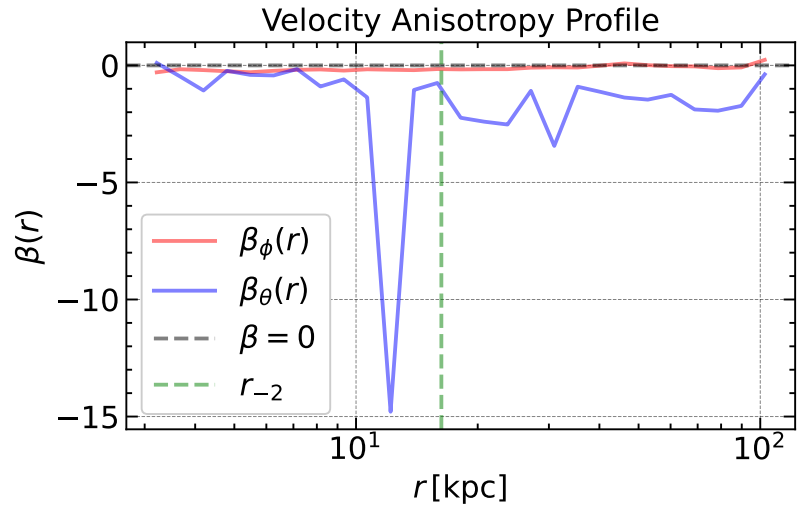
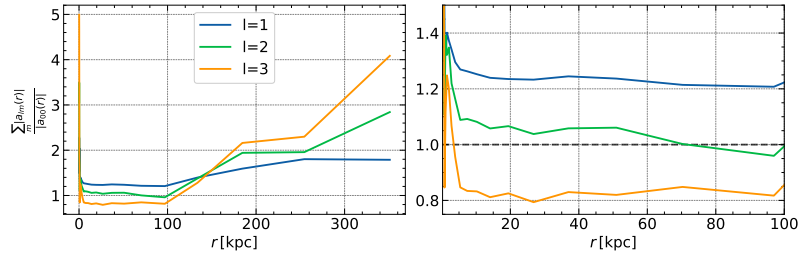
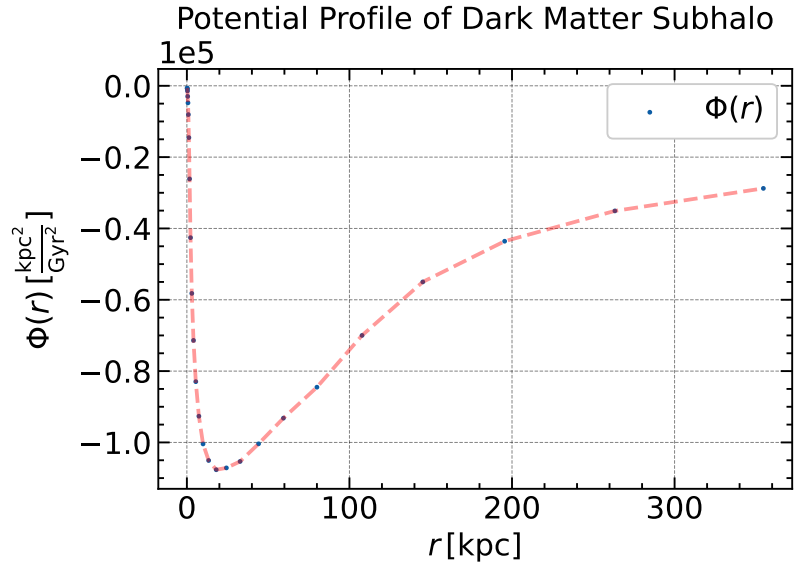












Chapter 3

Results and Discussion

Chapter 4

Conclusion

Appendix A

Appendix

A.0.1 Scripts

Bibliography